

# Shyam Sunder Singh

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## Education

Indian Institute of Technology (BHU), Varanasi  
Bachelor of Technology

CGPA: 8.26/10  
2024

## Work Experience

Associate Software Engineer – ITS Planners & Engineers, Hyderabad (June 2024 – Present)

### FAST (Flexible Adaptive Signal Timing) Algorithm:

- Developed **RESTful microservices** using **Python & FastAPI**, utilised **Pydantic** for input validation
- Reduced manual intervention by **60-80%** through automated model calibration using **Celery task queue**
- Leveraged **Polars/pandas pipelines** for data retrieval and automated module & API testing using **Pytest**
- Designed **gRPC/Redis** system to fetch and transmit traffic configurations and real-time states in **JSON**

### V2X (Vehicle-to-Everything):

- Engineered scalable **TCP socket server** in **Go**, handling **tens of thousands of concurrent connections**
- Achieved **80% faster data** transmission and reduced **DB connections by 10%** by integrating **Memcached**
- Architected task scheduler to deliver messages at **1Hz frequency** using **RQ Scheduler** in V2X systems

### TIM (Traffic Intelligence Module):

- Powering **650+ intersections** in **22+ cities** with scalable intelligent systems for real-time mobility control
- Reduced stops by **20-30%** at intersections by enabling **green corridor** via offset logic in coordinated signals
- Enhanced **operational stability** and reduced **transition glitches** by building a **sync mechanism**

Web Developer Intern – CyberCoderz Technologies (May 2023 – July 2023)

- Developed **4+ core website sections** with **React Router** and added client-side email support with **EmailJS**
- Leveraged **Git/GitHub** for version control and **collaborated** in an agile team to prototype and ship features

## Technical Skills

- **Languages:** Python, SQL, C/C++
- **Frameworks & Databases:** FastAPI, Celery, pandas, Polars, gRPC, MySQL, Redis, Memcached, Pytest
- **Cloud & Tools:** AWS (EC2, S3), Docker, Git, Linux, curl, Grafana, GitHub Actions
- **AI & API:** OpenAI API, WhatsApp Business API
- **Other:** Algorithms, Data Structures, Problem-Solving

## Projects

AI WhatsApp Bot — Period Tracker & Medical Articles || Message "Hi" to +91 81789 09407

- Built an AI WhatsApp chatbot tracking periods & sending alerts via **Redis, Celery, SQLAlchemy** on **AWS**
- Serves **26+ users daily**; integrated **OpenAI, WhatsApp API** for context-aware chat & S3 article delivery

School Management System (Ongoing) || Backend API || Frontend

- Developed the first module: **Student Attendance** System scaling to **200+ students/class** & **10k+ record**
- Implemented backend using **FastAPI** and **MySQL** with **SQLAlchemy** ORM and designed relational models

## Achievements & Certifications

- Certified **Postman API Fundamentals Student Expert**; proficient in API testing and usage [Certificate]
- Secured 3rd position and won a cash prize in Inter-IIT Civil Conclave, annual tech event at IIT Roorkee [Certi]
- Practiced **DSA and problem-solving**; Coding profiles links: LeetCode and GFG